

SettleMatch

Product manual

Reconcile Razorpay settlements to bank credits - safely.

What the product is

Every feature explained

Live UI walkthrough with screenshots

How to run, evaluate, and pitch

Sample batch: 90% match | 10 exceptions | 0 false matches

Rules first. AI second. Gate always.

github.com/kratagyaSingh-dev/SettleMatch

1 What SettleMatch is

SettleMatch is an AI Finance Controller for Razorpay AI Buildathon Track 04. It closes one real finance-ops loop: take a Razorpay settlement export and a bank statement, match every rupee that can be proven, and refuse every rupee that cannot.

The problem

- Money arrives in two lists: Razorpay settlements and bank credits.
- Formats differ (PDF, Excel, Word, CSV, TXT) and narrations are messy.
- Finance still matches these in Excel - slow, error-prone, hard to audit.
- A generic chatbot guesses links. Finance needs verification, not generation.

The promise

- Rules close clear cases first (UTR, amount + date).
- Gemini sees leftovers only, and only from a short candidate list.
- Confidence gate (default 0.85) turns weak AI picks into exceptions.
- Collisions and already-used bank rows are blocked - no double-spend.
- Humans review the exception queue only. They do not re-check every row.

2 Pipeline

```
Documents (PDF / Excel / Word / CSV / TXT)
  -> Extract strict columns only
  -> Rules: exact UTR, then unique amount +/- 1 day
  -> Collision detector
  -> Gemini on leftovers (candidate list only)
  -> Confidence gate (>= 0.85, no rematch)
  -> Matches + explanations | Exceptions + review
  -> Dashboard / Simulator / PDF / Word / audit ZIP
```

Stage	Behavior	If it fails
Extract	Map required columns only	Hard error on empty/bad file
Rules	UTR exact; amount + date	Multi-candidate = collision
AI	JSON pick from candidates	Invalid ID / quota = refuse
Gate	Threshold + used-bank check	Below 0.85 = exception
Review	Approve / Reject / Follow-up	Does not auto-move money

3 Sample batch results (localhost data)

These numbers come from uploading data/samples/settlements.csv (100 rows) and data/samples/bank.csv (125 rows), then clicking Reconcile.

Metric	Value
Settlements / bank rows	100 / 125
Match rate	90.0% (90 matched)
Rules / AI	75 / 15
Exceptions (honest refusals)	10
Collisions blocked	15
Money matched	Rs 330,755
Money at risk	Rs 47,774
Recovery rate	87.4%
Eval precision / recall	100% / 100%
False matches	0

4 Every feature explained

4.1 Multi-format extract

Users do not convert files first. Drop PDF, Excel, Word, CSV, or TXT. Any mix works (settlements PDF + bank Excel is valid). Only required columns are kept.

- Settlements (7): settlement_id, payment_id, amount, currency, utr, settled_at, status
- Bank (5): bank_txn_id, amount, narration, value_date, utr
- Aliases work (UTR No, settlement date, particulars). Extra columns are dropped.
- After upload, a green preview shows filename, row count, and format.

4.2 Upload and Reconcile

Upload is the live demo path. Two uploaders, a confidence-gate slider (0.50-0.99, default 0.85), and Reconcile. The button stays disabled until both files extract cleanly. On success the app jumps to Dashboard. Reset clears matches, exceptions, reviews, and simulator.

4.3 Rules engine

Rules run before any AI call. They are cheap, explainable, and high precision.

Rule	When it fires	Confidence
Exact UTR	Same UTR + amount within Rs 0.01	1.00
Amount + date	Unique amount inside +/- 1 day	1.00

On this batch, rules close 75 of 100 settlements with zero AI cost.

4.4 Collision detection

Two similar credits and one settlement is how bad tools invent money. SettleMatch blocks duplicate UTRs, amount windows with more than one unused bank row, and already-used bank_txn_id values. Ambiguous cases are logged, never auto-approved.

4.5 Gemini on leftovers only

AI is not the first matcher. It sees one leftover settlement plus a short candidate list. It must return JSON (bank_txn_id, confidence, reason, refuse) and cannot invent an ID outside that list. If GEMINI_API_KEY is missing or quota fails, a heuristic fallback still finishes the loop so the demo never dies.

4.6 Confidence gate

Every AI suggestion is refused if refuse=true, bank_txn_id is null, confidence is below the slider, or the bank row is already matched. Refused rows become exceptions with a written reason. AI can suggest. It cannot close the books alone.

4.7 Explainability

Every accepted match stores stage, method (rule_exact_utr / rule_amount_date / ai_gated), confidence, a summary, and a step list. On Matches, pick any settlement_id and read the trail. Exceptions get the same treatment: why refused, Rs at risk, next human action.

4.8 Exception queue and human review

This batch has 10 honest refusals - not a fake 100% score. Finance can Approve manually, Reject / keep open, or mark Needs finance follow-up. Reviews stay in the workspace. They do not auto-move money.

4.9 Dashboard

Four KPIs: match rate, recovery rate, money matched, money at risk. Plus rules/AI split, collisions blocked, gate used, and a ground-truth eval strip when output/eval_metrics.json exists. Charts: match breakdown, close-rate donut, money recovered vs at risk, rules vs AI.

4.10 Gate simulator

Re-runs the same files across thresholds. Tight gate (0.90+) = fewer matches, cleaner books. Loose gate (0.70) = more matches, more risk. Default 0.85 is the evaluated balance.

4.11 Export and audit pack

Download	What you get
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PDF report	Stats, matches, exceptions, source filenames
Word report	Same content, editable for finance notes
Audit ZIP	CSVs + collisions + audit_log.jsonl + report.json
Matches CSV	Spreadsheet import

4.12 Connections (how this scales)

Manual upload does not scale to millions of users. Connections is the production path: Razorpay test-mode Key ID, bank SFTP or webhook, daily 02:00 IST schedule, Auto-run toggle, last/next run cards, and a recent auto-runs table. Same engine as Upload. Humans only touch the exception queue.

4.13 Ground-truth eval

python eval/run_eval.py compares output to data/expected/ground_truth.csv (correct / false / missed matches, true exceptions respected, precision and recall). audit_extract.py covers 25 format combinations. smoke_test.py checks the happy path.

5 How to run (before the walkthrough)

1. Open a terminal in the SettleMatch folder.
2. Activate the venv: `.\venv\Scripts\Activate.ps1`
3. Install if needed: `pip install -r requirements.txt`
4. Start the app: `streamlit run app.py`
5. Open `http://localhost:8501`
6. Keep `data/samples/settlements.csv` and `data/samples/bank.csv` ready.

6 Walkthrough - Upload

1. Sidebar -> Upload. This is the starting page.
2. Upload Settlements export, then Bank statement (PDF / Excel / Word / CSV / TXT).
3. Green previews appear: 100 rows + 125 rows on the sample files.
4. Leave the confidence gate at 0.85 unless you want a tighter refuse bar.
5. Click Reconcile. Pipeline runs extract -> rules -> AI -> gate -> audit.
6. On success the app opens Dashboard automatically.

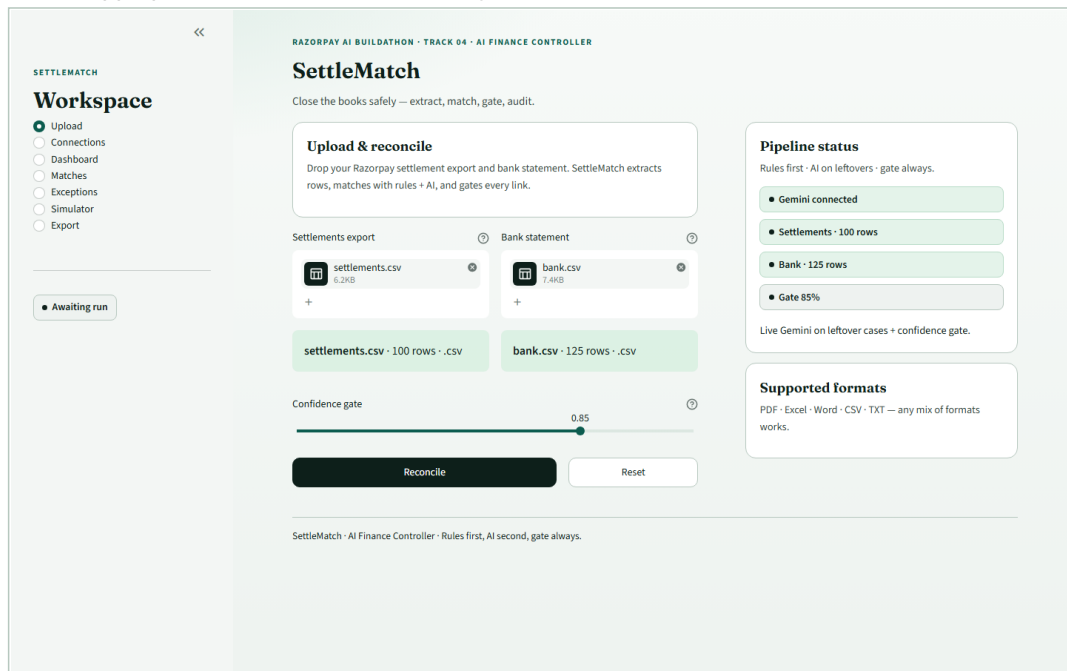


Fig 6 - Upload with both sample files extracted. Reconcile is enabled.

7 Walkthrough - Connections

1. Sidebar -> Connections. This is the production story, not a second matcher.
2. Razorpay test Key ID + masked secret = nightly settlements API pull.
3. Bank ingest: SFTP path or webhook URL.
4. Schedule Daily 02:00 IST and keep Auto-run ON.
5. Pitch line: companies connect once; finance only reviews exceptions.

SETTLEMATCH

Workspace

- ☐ Upload
- ☒ Connections
- ☐ Dashboard
- ☐ Matches
- ☐ Exceptions
- ☐ Simulator
- ☐ Export

● **Awaiting run**

RAZORPAY AI BUILDATHON · TRACK 04 · AI FINANCE CONTROLLER

SettleMatch

Close the books safely — extract, match, gate, audit.

Connections

Production ingest — Razorpay API + bank SFTP/webhook on a daily schedule. Upload tab stays for ad-hoc runs; this is how finance scales.

Data sources

Connect once per merchant — nightly jobs pull fresh rows automatically.

Razorpay (test mode)

Key ID:

Key secret:

Bank ingest

Ingest method: ☒ SFTP ☐ Webhook

SFTP path:

Schedule

Hour: Minute: ☒ Auto-run

Live status

What runs while your team sleeps.

- Razorpay - test mode connected
- SFTP - sftp.hdfcbank.com/settlematch...
- Daily 02:00 IST
- Auto reconcile - ON

LAST RUN

Auto ✓
23 Aug 2026, 02:00 AM

NEXT RUN

Scheduled
24 Aug 2026, 02:00 AM

Fig 7 - Connections: Razorpay test API, bank SFTP, schedule, last auto-run.

8 Walkthrough - Dashboard

1. After Reconcile, Dashboard is the reviewer screen.
2. KPIs on this batch: match 90.0%, recovery 87.4%, matched Rs 330,755, at risk Rs 47,774.
3. Split: 75 rules + 15 AI. 10 exceptions listed honestly.
4. Read the charts, then the eval strip: precision/recall 100%, 0 false matches.

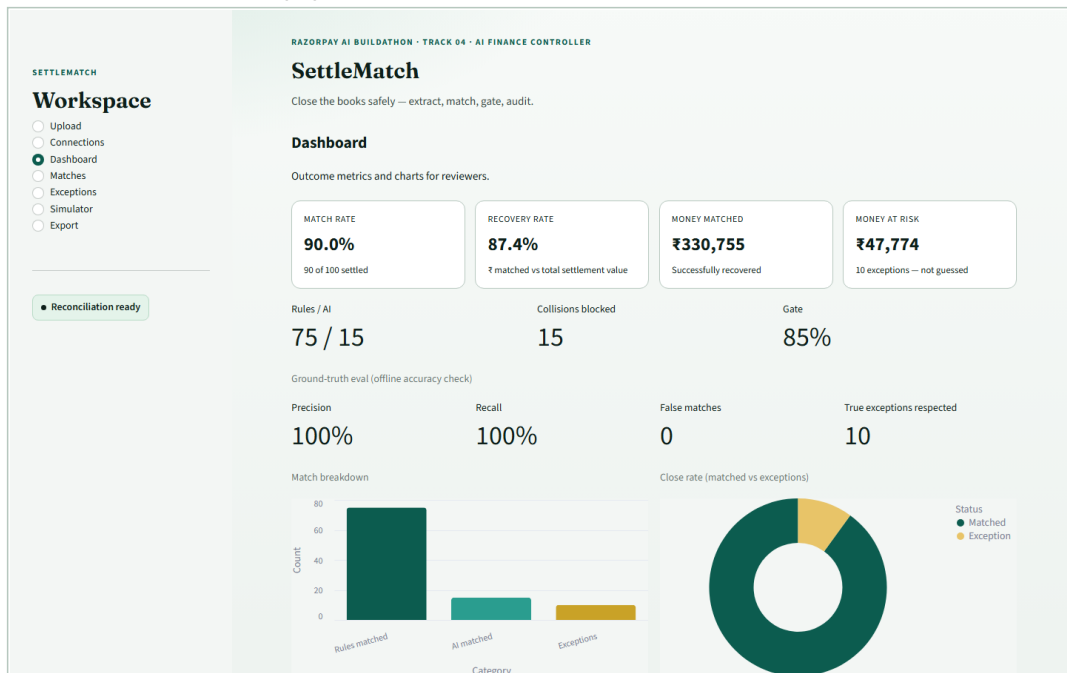


Fig 8 - Dashboard after the live run (90 matched, 10 exceptions).

9 Walkthrough - Matches

1. Sidebar -> Matches. All accepted pairings are in the table.
2. Columns include settlement_id, bank_txn_id, method, confidence.
3. Select any settlement in the dropdown to open the explanation panel.
4. You should see stage, method, confidence, summary, and the step list.

5. Pitch point: every rupee link is explainable. Nothing is a black box.

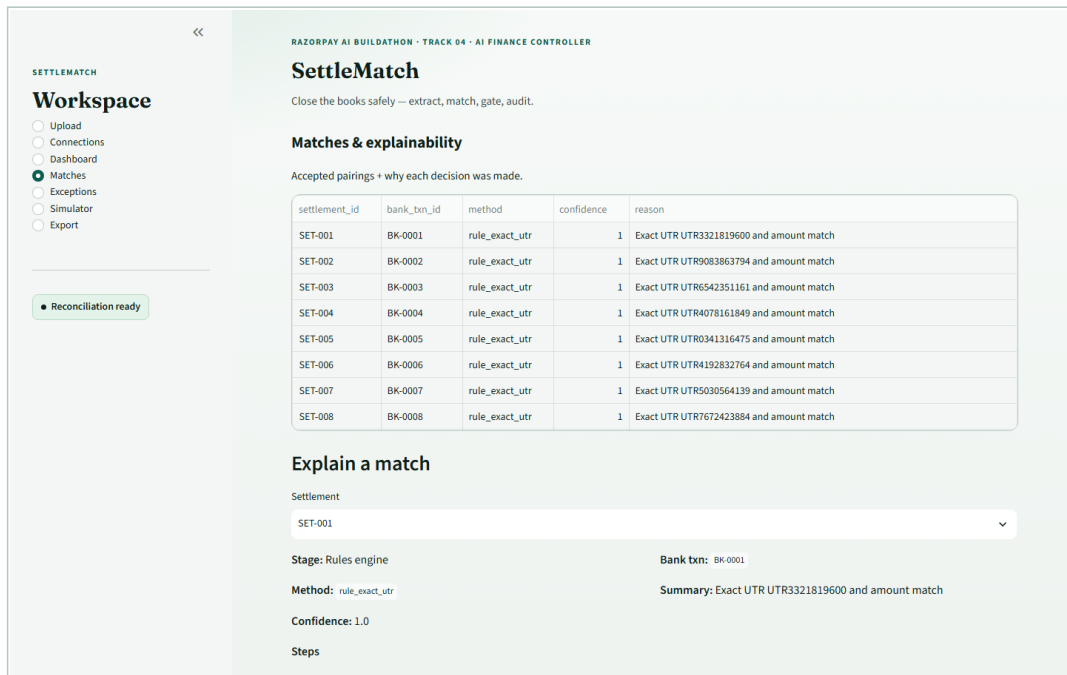


Fig 9 - Matches table plus step-by-step explanation for one settlement.

10 Walkthrough - Exceptions

1. Sidebar -> Exceptions. You will see 10 refused rows on this batch.
2. Each row has a reason: low confidence, collision, or no safe candidate.
3. The amount chart shows rupees still at risk.
4. Pick a row. Use Approve manually, Reject / keep open, or Needs follow-up.
5. Finance works only here. The rest of the batch is already closed.

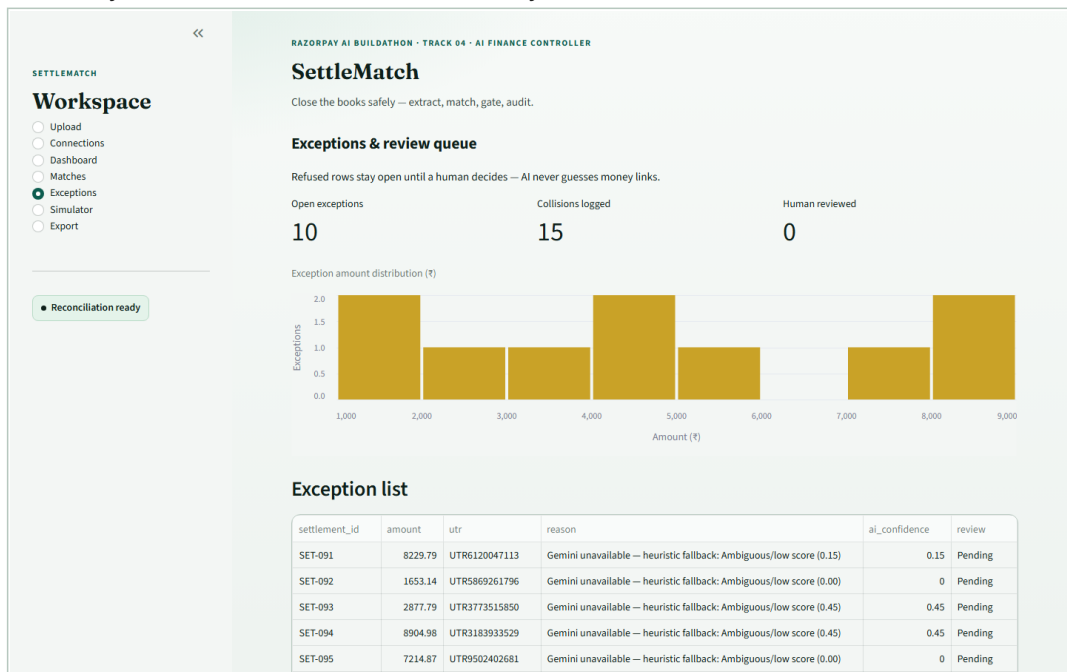


Fig 10 - Exception queue with reasons, amount chart, and review actions.

11 Walkthrough - Simulator

1. Sidebar -> Simulator. This uses the same uploaded files, not empty state.
2. The line chart shows match rate as the gate moves from 0.50 to 0.99.

- The right chart shows how exceptions grow as the gate tightens.
- Use 0.85 as the default. Show 0.70 vs 0.90 if a reviewer asks about risk.

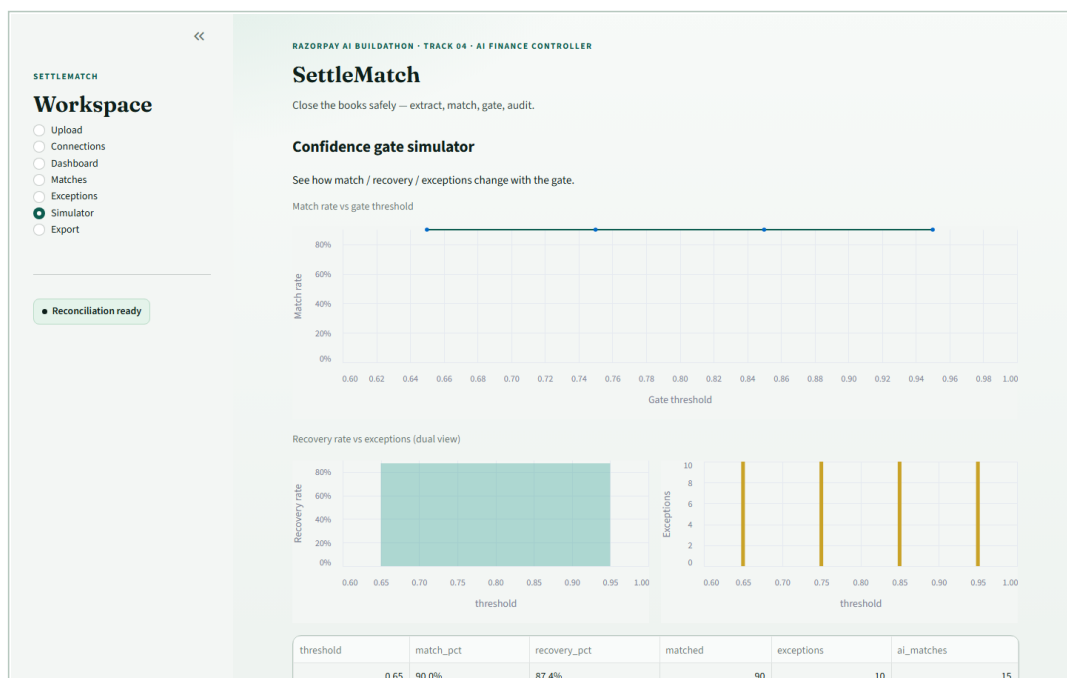


Fig 11 - Gate simulator after reconciliation (not the empty-state page).

12 Walkthrough - Export

- Sidebar -> Export.
- Download PDF report for a one-pager, Word if finance will annotate.
- Download Audit pack ZIP for the full trail: matches, exceptions, collisions, audit log.
- Matches CSV is for spreadsheet import.
- Scroll the audit table if a reviewer asks why one ID was accepted or refused.

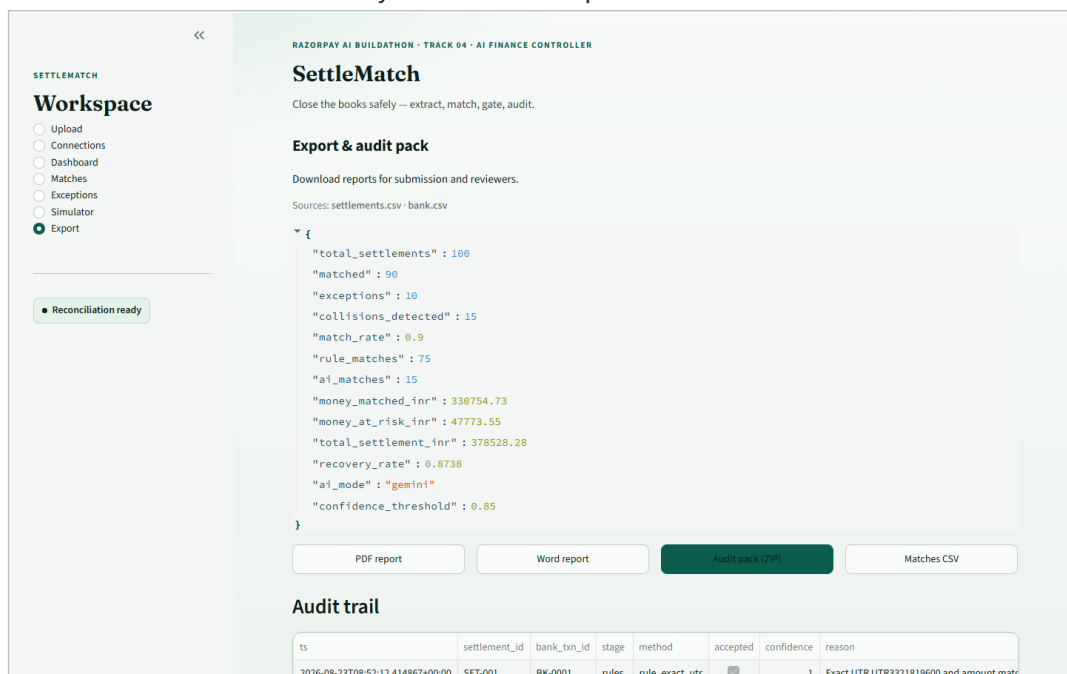


Fig 12 - Export downloads and the full audit trail.

13 Copy-paste setup

```
git clone https://github.com/kratagyaSingh-dev/SettleMatch.git
cd SettleMatch
python -m venv .venv
.\.venv\Scripts\Activate.ps1
pip install -r requirements.txt
copy .env.example .env
streamlit run app.py
# http://localhost:8501 -> Upload both files -> Reconcile
```

Add GEMINI_API_KEY in .env for live Gemini. Without it, the heuristic matcher still runs the full loop. Never commit .env.

14 Tech stack

Layer	Technology
Language / UI	Python 3 + Streamlit
Data / charts	Pandas + Altair
AI	Gemini 3.6 Flash + heuristic fallback
Extract	pdfplumber, openpyxl, python-docx
Reports	fpdf2, python-docx, ZIP
Config	python-dotenv

Not in this demo, by design: React, PostgreSQL, Docker, live Razorpay API. Those belong on the Connections production path.

15 Repository map

Path	Purpose
app.py	7-page Streamlit product UI
src/extract.py	Multi-format extract + column map
src/rules.py	UTR and amount/date matcher
src/collisions.py	Ambiguity / double-spend checks
src/ai_matcher.py	Gemini + heuristic fallback
src/gate.py	Confidence threshold gate
src/reconcile.py	Pipeline orchestrator
src/export_report.py	PDF / Word / ZIP
eval/run_eval.py	Ground-truth accuracy
data/samples/	Upload files used in this manual
architecture.md	System design for reviewers

16 Safety invariants

- AI never invents a bank_txn_id outside the candidate set.
- Already-matched bank rows cannot be rematched.
- Ambiguous collisions are refused, not guessed.
- Below-threshold AI suggestions become exceptions.
- Every accept and refuse is written to the audit log.
- Human review never silently moves money.

17 Five-minute pitch

1. 0:00 Problem: two lists, manual matching, audit pain.
2. 0:45 Upload live files. Click Reconcile.
3. 1:30 Dashboard: 90% match, Rs 330,755 recovered, 0 false matches.
4. 2:30 One exception + human review. We refuse. We do not guess.
5. 3:30 Connections: nightly Razorpay + bank ingest.
6. 4:30 Download the audit pack. Close.

Spoken closer: SettleMatch does not just find matches. It closes the books - with proof.